THE TANGENT LINE AT A MARKED POINT ON $M_{0,n}/\mathfrak{S}_{n-1}$, AFTER A TABLE OF GROTHENDIECK

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Abstract. Let L_n be the line bundle on $M_{0,n}$ whose fibre is the tangent line to $\mathbb{P}^1$ at the last marked point, and L the tensor product of the tangent lines at all of them. We show that, in characteristic zero, the Picard group of the quotient stack $[M_{0,n}/\mathfrak{S}_{n-1}]$ is cyclic of order $(n-1)(n-2)$ for every $n \geq 3$, generated by L_n, and that L has order $n-1$ in $\mathrm{Pic}[M_{0,n}/\mathfrak{S}_n]$. For even $n \geq 6$ the group was known, through the integral Chow ring computed by Edidin and Hu; the odd case does not seem to be in print. The question comes from a table in Grothendieck's working notes, which gives these orders for four points and only guesses for five and six. The proof is a few lines once the stack is written as monic squarefree polynomials modulo the affine group. An elementary version, with explicit trivialisations and fixed configurations, is formalised in Lean.

1. Introduction

Let k be a field of characteristic zero and $n \geq 3$. Let $M_{0,n}$ be the moduli space of n distinct ordered points $x_1, \ldots, x_n$ on $\mathbb{P}^1$, up to PGL_2. For each i the tangent lines $T_{x_i}\mathbb{P}^1$ form a line bundle L_i on $M_{0,n}$, and we put $L = L_1 \otimes \cdots \otimes L_n$. The symmetric group $\mathfrak{S}_n$ acts on $M_{0,n}$ by renumbering the points and acts compatibly on L; the subgroup $\mathfrak{S}_{n-1}$ fixing the last point acts on L_n. This defines classes

$$[L] \in \mathrm{Pic}[M_{0,n}/\mathfrak{S}_n], \qquad [L_n] \in \mathrm{Pic}[M_{0,n}/\mathfrak{S}_{n-1}],$$

and the question is their order: the least $m \geq 1$ for which $L^{\otimes m}$, or $L_n^{\otimes m}$, has a nowhere-vanishing section compatible with the action.

Theorem 1. *For every $n \geq 3$:*

(1) $\mathrm{Pic}[M_{0,n}/\mathfrak{S}_{n-1}]$ *is cyclic of order $(n-1)(n-2)$, generated by $[L_n]$;*
(2) $[L]$ *has order $n-1$ in* $\mathrm{Pic}[M_{0,n}/\mathfrak{S}_n]$.

For $n = 4, 5, 6$ the orders are $3, 4, 5$ for L and $6, 12, 20$ for L_n.

The source. The University of Montpellier holds the mathematical papers that Alexandre Grothendieck left to Jean Malgoire, and publishes their facsimiles. Its folder 146, on the action of symmetric groups on $M_{0,n}$, has a table on page 70 giving, for $n = 4, 5, 6$, the least exponent that makes L and the L_i trivial compatibly with the symmetric group. The column $n = 4$ reads 3 for L and 6 for L_i, which is right. The columns $n = 5$ and $n = 6$ carry only queried values: *12 ?* for both bundles when $n = 5$, and *20 ?* for L and *60 ?* for L_i when $n = 6$. Page 72 builds sections indexed by the three-element subsets of the points and concludes that $L^{\otimes 6}$ is trivial for $n = 4$ and $n = 5$. Theorem 1 fills in the table: for $n = 5$ the queried 12 is right for L_5 but not minimal for L, whose order is 4; for $n = 6$ the orders are 5 and 20, not 20 and 60.

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	$n=4$		$n=5$		$n=6$	
	L	L_n	L	L_n	L	L_n
Page 70	3	6	12 ?	12 ?	20 ?	60 ?
Theorem 1	3	6	4	12	5	20

TABLE 1. The table of folder 146, page 70, and the orders given by Theorem 1.

What was known. The stack $[M_{0,n}/\mathfrak{S}_n]$ is the stack of smooth binary forms of degree n up to scalars. Its Picard group was computed by Arsie and Vistoli [1] and, for n even, by Gorchinskiy and Viviani, who find $\mathbb{Z}/2(n-1)$ [5, Th. 3.6]; the order of $[L]$ follows from their methods in a few lines, but we have not found it stated. For $n = 2g+2$ even and $g \geq 2$, Edidin and Hu compute the integral Chow ring of $[M_{0,2g+2}/\mathfrak{S}_{2g+1}]$, which they present as binary forms with one marked root, as $\mathbb{Z}[l]/(2g(2g+1)\,l)$ [3, Th. 1.3], building on Pernice's description [7]; so $\mathrm{Pic}[M_{0,n}/\mathfrak{S}_{n-1}]$ is cyclic of order $(n-1)(n-2)$ for even $n \geq 6$. Their generator is the class of a character of a torus, and the identification with L_n is not stated. For $n = 4$, the stack $\mathcal{M}_{1,1}$ is a μ_2-gerbe over $[M_{0,4}/\mathfrak{S}_3]$, and $\mathrm{Pic}\,\mathcal{M}_{1,1} = \mathbb{Z}/12$ [6, 4] gives the order 6. For odd n we found no printed statement; D. Edidin confirms that their paper treats only the even case and that he knows no reference for the odd one [2]. The question was asked on MathOverflow by the author (question 516243), with the argument of Section 2 as a self-answer.

2. The stack as polynomials modulo the affine group

Put $m = n-1 \geq 2$. Since PGL_2 acts transitively on $\mathbb{P}^1$, we may send the last point to ∞; its stabiliser is the affine group $\mathrm{Aff} = \mathbb{G}_a \rtimes \mathbb{G}_m$, acting on $\mathbb{A}^1$ by $x \mapsto ax+b$, and the other m points, unordered, are the roots of a monic squarefree polynomial of degree m. Taking the slice $x_n = \infty$ gives an isomorphism of stacks

$$[M_{0,n}/\mathfrak{S}_{n-1}] \;\cong\; [U_m/\mathrm{Aff}],$$

where $U_m \subset \mathbb{A}^m$ is the open set of monic squarefree polynomials $x^m + c_1 x^{m-1} + \cdots + c_m$, on which Aff acts by substitution.

The translations $\mathbb{G}_a$ act freely, and every orbit meets the closed set $c_1 = 0$ exactly once, since m is invertible in k: the translation $x \mapsto x - c_1/m$ centres the roots. So

$$[U_m/\mathrm{Aff}] \;\cong\; [V_m/\mathbb{G}_m], \qquad V_m = \{x^m + c_2 x^{m-2} + \cdots + c_m \text{ squarefree}\} \subset \mathbb{A}^{m-1},$$

where $\lambda \in \mathbb{G}_m$ acts by $x \mapsto \lambda x$, that is, on the coefficient c_j with weight j. The complement of V_m in $\mathbb{A}^{m-1}$ is the discriminant hypersurface $\mathrm{disc} = 0$. The discriminant of the generic centred polynomial is irreducible, and as a function of the roots it is $\prod_{i<j}(r_i - r_j)^2$, homogeneous of degree $m(m-1)$; so it is a semi-invariant of weight $m(m-1)$.

Lemma 2. $\mathrm{Pic}[V_m/\mathbb{G}_m] \cong \mathbb{Z}/m(m-1)$, *generated by the line bundle attached to the character* $\lambda \mapsto \lambda$.

Proof. The equivariant Picard group of the affine space $\mathbb{A}^{m-1}$ is the character group $\mathbb{Z}$ of $\mathbb{G}_m$. Removing the irreducible divisor $\mathrm{disc} = 0$ gives the exact sequence

$$\mathbb{Z} \cdot [\mathrm{disc} = 0] \longrightarrow \mathrm{Pic}^{\mathbb{G}_m}(\mathbb{A}^{m-1}) = \mathbb{Z} \longrightarrow \mathrm{Pic}^{\mathbb{G}_m}(V_m) \longrightarrow 0,$$

and the class of the divisor of a semi-invariant of weight w is the character of weight w, here $w = m(m-1)$. □

Proof of Theorem 1. In the model $[U_m/\mathrm{Aff}]$ the last point is ∞. In the coordinate $u = 1/x$ at ∞, the map $x \mapsto ax+b$ is $u \mapsto u/(a+bu)$, whose derivative at $u = 0$ is a^{-1}; so $\mathbb{G}_m$ acts on $T_\infty \mathbb{P}^1$ through the character of weight -1, and $[L_n]$ corresponds to $-1 \in \mathbb{Z}/m(m-1)$. It generates the group, which proves (1).

At a finite root r, the derivative of $x \mapsto ax + b$ is a, so each $T_{r_i}\mathbb{P}^1$ has weight $+1$, and the pull-back of L to $[M_{0,n}/\mathfrak{S}_{n-1}]$ has weight $m - 1 = n - 2$. Its order in $\mathbb{Z}/(n-1)(n-2)$ is $n-1$, so the order of $[L]$ is a multiple of $n-1$. Conversely, write the points as vectors $v_i \in k^2 \smallsetminus \{0\}$ and $[v_i v_j] = \det(v_i, v_j)$. Since $T_{\langle v\rangle}\mathbb{P}^1 \cong \mathrm{Hom}(\langle v\rangle, k^2/\langle v\rangle) \cong \langle v\rangle^{\otimes -2} \otimes \det k^2$, trivialising $L^{\otimes(n-1)}$ compatibly with the action amounts to giving a nowhere-vanishing function of the v_i, homogeneous of degree $2(n-1)$ in each v_i, transforming under $g \in \mathrm{GL}_2$ by $\det(g)^{n(n-1)}$ and invariant under renumbering. The function

$$\prod_{i \neq j} [v_i v_j] \;=\; (-1)^{n(n-1)/2} \prod_{i<j} [v_i v_j]^2$$

is such a function: each v_i occurs in $2(n-1)$ brackets, it vanishes nowhere on $M_{0,n}$, and the right-hand side is invariant under $\mathfrak{S}_n$. This proves (2). □

Remark 3. For $n = 3$ the stack $[M_{0,3}/\mathfrak{S}_2]$ is $B(\mathbb{Z}/2)$, with Picard group $\mathbb{Z}/2$, as the theorem says. The same computation applied to $[M_{0,n}/\mathfrak{S}_n]$, binary forms modulo GL_2, recovers the results of Arsie–Vistoli and Gorchinskiy–Viviani.

3. An elementary version

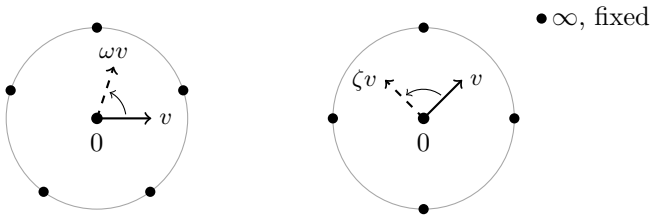
The argument can be phrased without stacks, in the vector model used above. A trivialisation of $L_n^{\otimes m}$ compatible with $\mathfrak{S}_{n-1}$ is, in the same way, a nowhere-vanishing function $F(v_1, \dots, v_n)$, homogeneous of degree $2m$ in v_n and 0 in the other v_i, transforming by $\det(g)^m$ under GL_2, and symmetric in $v_1, \dots, v_{n-1}$.

Upper bound. The function

$$F = \Big(\prod_{i<n} [v_i v_n]\Big)^{2(n-2)} \Big/ \prod_{\substack{i \neq j \\ i,j<n}} [v_i v_j]$$

has these properties for $m = (n-1)(n-2)$.

Lower bound. Let ω be a primitive $(n-1)$-th root of unity, and consider the configuration with $x_n = 0$ and $x_1, \dots, x_{n-1}$ the $(n-1)$-th roots of unity. Write $0 = \langle(0,1)\rangle$ and the other points as $\langle(z,1)\rangle$. The matrix $g = \mathrm{diag}(\omega, 1)$ fixes $(0,1)$ and permutes the vectors $(z,1)$, so $F(gv) = F(v)$ by symmetry, while $F(gv) = \det(g)^m F(v) = \omega^m F(v)$; hence $\omega^m = 1$. With ζ a primitive $(n-2)$-th root of unity and the configuration $0, \infty$ and the $(n-2)$-th roots of unity, the same argument with $\mathrm{diag}(\zeta, 1)$ gives $\zeta^m = 1$; it multiplies the vector $(1,0)$ of ∞ by ζ, which F does not see, being of degree 0 in that variable. As $n-1$ and $n-2$ are coprime, $(n-1)(n-2)$ divides m (Figure 1).



0 and μ_5, rotated by $z \mapsto \omega z$ ∞ and μ_4, rotated by $z \mapsto \zeta z$

FIGURE 1. The two configurations of the lower bound for $n = 6$. The last point is 0; a rotation permutes the other points and turns the tangent vector v at 0 into ωv or ζv. A trivialisation compatible with the renumbering must take the same value at both, so $5 \mid m$ and $4 \mid m$, hence $20 \mid m$ for L_6.

Both directions, for every n, and the corresponding statements for L, are formalised in Lean 4 with Mathlib, over any field containing the relevant roots of unity (theorems `trivL_iff` and `trivLi_iff` of `Grothendieck.Folder146`, in the repositories Commutator-IO/grothendieck-project and Commutator-IO/grothendieck-project-palomar). What is not

formalised is the dictionary between equivariant line bundles on $M_{0,n}$ and such functions, that is, the identification of the equivariant Picard group with the first cohomology of $\mathfrak{S}$ with values in the units; Mathlib does not have the spaces $M_{0,n}$.

Statement on the use of artificial intelligence

The transcription of folder 146, the search of the literature, the general formula, its proofs, the Lean formalisation and a first draft of this text were produced with language models (Claude Opus 5 and Opus 5.5, Anthropic), under the author's direction. The author checked the proofs and the references, and takes full responsibility for the content. The formal proofs are checked mechanically by Lean.

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